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Abstract

Vapor compression distillation (VCD) can reduce the energy use and cost of magnesium distillation refining by up to 90% versus today's batch distillation processes. This work describes a new continuous gravity-driven multiple effect thermal system (G-METS) process for magnesium VCD with just one internal moving part. The distiller will likely use less than 1 kWh/kg magnesium product, and high throughput of the continuous process can lower capital cost considerably. A detailed thermal model of the system presented here describes multicomponent evaporation, and batch distiller experiments validate key components of the model. There are multiple alloy distillation challenges with potential solutions described here, including liquid diffusion resistance, aerosol carry-over, and removal of volatile elements such as zinc. This efficient low-cost process could play a key role in multiple new flow sheets, from magnesium alloy recycling to rare earth magnet recycling to primary magnesium production.

Keywords Magnesium - Recycling and secondary recovery - Modeling and simulation - Distillation

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Efficient Low-Cost Gravity-Driven Multiple Effect Thermal System (G-METS) Distillation of Magnesium

Armaghan Ehsani Telgerafchi, Gabriel Espinosa, Madison Rutherford, Adam Powell, and David Dussault

Abstract

Vapor compression distillation (VCD) can reduce the energy use and cost of magnesium distillation refining by up to 90% versus today's batch distillation processes. This work describes a new continuous gravity-driven multiple effect thermal system (G-METS) process for magnesium VCD with just one internal moving part. The distiller will likely use less than 1 kWh/kg magnesium product, and high throughput of the continuous process can lower capital cost considerably. A detailed thermal model of the system presented here describes multicomponent evaporation, and batch distiller experiments validate key components of the model. There are multiple alloy distillation challenges with potential solutions described here, including liquid diffusion resistance, aerosol carry-over, and removal of volatile elements such as zinc. This efficient low-cost process could play a key role in multiple new flow sheets, from magnesium alloy recycling to rare earth magnet recycling to primary magnesium production.

Keywords

Magnesium • Recycling and secondary recovery • Modeling and simulation • Distillation

Introduction

Magnesium (Mg) is a low-density metal whose alloys have outstanding stiffness/weight and excellent strength/weight. It is proposed for use in very lightweight high-efficiency motor vehicles. For example, in 2015 Ford and Magna demonstrated an aluminum-intensive vehicle, which they called "Mach 1", with 25% weight reduction vs. a steel baseline. Their magnesium-intensive "Mach 2" vehicle design achieved 50% lower weight, though in practice would probably be 40–45% lighter than the steel baseline [1]. Two of the barriers to more widespread magnesium use are: low-cost low-emissions primary production method suitable for OECD or similar labor and environmental regulatory framework for supply and cost stability; and a recycling method capable of producing low-iron low-copper low-nickel magnesium from post-consumer scrap.

Magnesium metal distillation can address both of these barriers, enabled by its relatively low boiling point of 1091 °C —indeed, about 15% of the more than 2000 patents in US classification C22B26/22 "Obtaining magnesium" involve distillation. For recycling, it can use post-consumer scrap and produce pure magnesium with copper and nickel concentrations below even the demanding specifications of low-corrosion alloys such as AZ91E [2]. For primary production, distillation enables direct reduction of magnesium oxide (MgO) in a Hall–Héroult type cell with a dense reactive liquid metal cathode such as tin to make an alloy, then separation of pure magnesium from the alloy and returning the host metal to the electrolysis cell [3, 4], as discussed further in a separate paper in these Proceedings. However, the high vapor pressure of magnesium at its melting point makes it challenging to condense as a liquid, particularly when partially diluted by an inert gas [5–7]. In addition, high temperature generally leads to low energy efficiency, with distillers using 5–7 kWh/kg Mg product, cf. 1.5 kWh/kg enthalpy of vaporization.

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Vapor compression distillation uses the heat released in condensation to provide much of the heat required for evaporation, and can dramatically reduce the energy use and cost of distillation. Pumping magnesium vapor at high mass flow rate is very challenging, but multiple effect thermal system (METS) distillers overcome this by pumping liquid rather than vapor [8–12], and there are entire patent classifications based on this principle (in USA: B01D1/26 "Multiple effect evaporating", B01D1/0088 "Cascade evaporators"). Either way, each condenser operates at higher pressure than an adjacent evaporator, such that the boiling point is higher in the condenser, and heat flows from the hotter condenser to the cooler evaporator.

Here we present a new METS magnesium distiller concept which uses gravity to create this pressure difference with no moving parts, called G-METS. The weight of a column of liquid metal raises the pressure in the evaporating chamber below it, and thus raises the pressure in the condenser connected with the lower evaporating chamber, as shown in Fig. 1. Heat flows upward from the main heater, and each condenser feeds the heat needed for the evaporator above it. Standpipes control liquid level in each evaporator. This is similar to the water distiller of Petrek [10], which uses vertical drain pipes between effects to achieve the desired pressure difference. But in this magnesium distiller, a barrier and tube separate each evaporator from the condenser in the same effect, leading to better thermal efficiency and segregation of volatiles such as zinc. Other features can potentially further enhance zinc separation.

This work first describes the G-METS distiller, and features which overcome specific challenges. We then present:

- Results of steel corrosion tests and one- and two-effect batch distiller experiments;
- A reduced-order thermal model which estimates temperatures, pressures, compositions, and flow rates;
- Solid modeling of an industrial distiller design which illustrates the distiller scale, and structural Finite Element Analysis (FEA) using the solid model estimates maximum stress;
- FEA with fluid flow and heat transfer estimates vapor velocity distribution in the evaporator and condenser.

Distiller Description

Figure 1 shows a schematic of a four-effect magnesium G-METS distiller. For recycling, solid alloy is introduced into the top melter. Melted Mg alloy flows at a controlled rate downward into the top evaporator, where a portion of the Mg alloy evaporates. That portion which evaporates rises to the top condenser, where it condenses. The remaining magnesium which does not evaporate here flows down through the liquid standpipe into the next evaporator, where a portion of it evaporates, and so on. Each evaporator-condenser pair is called an "effect". Heat flows upward from effect to effect, from lower condensers to the evaporators above them. Liquid Mg flows downward between evaporators from effect to effect. Mg vapor flows upward within each effect from its evaporator into its condenser above it. Standpipes control the height of liquid Mg alloy in each evaporator.

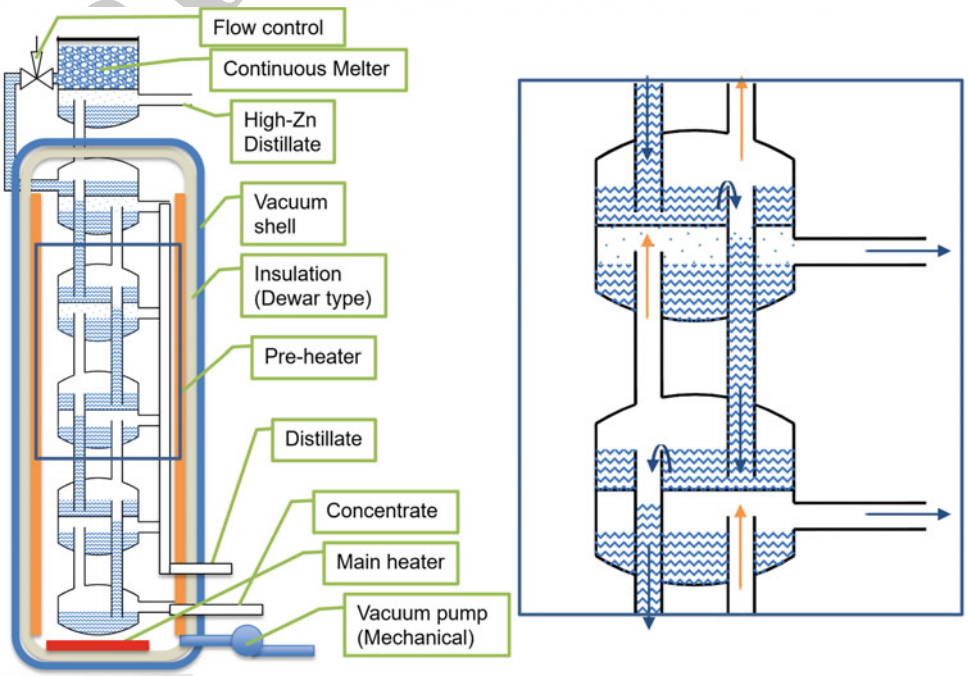


Fig. 1 Schematic diagram of a four-effect magnesium G-METS distiller, with detailed view of a single effect

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Several issues complicate the operation of a G-METS distiller. First, if the main heater is run at constant power, the distiller is unstable, in that more liquid metal flow downward from evaporator to evaporator leads to lower temperatures, reducing evaporation rate, and increasing liquid metal flow rate further. Using a proportional integral derivative (PID) or similar algorithm to control temperature in the bottom evaporator avoids this instability: faster flow leads to higher heat input and evaporation rate, slowing down concentrate flow. One can further enhance stability by regulating flow of influent metal to control the temperature at the top of the distiller.

Second, in recycling it is necessary to heat and melt the incoming magnesium alloy metal, which uses energy. This distiller can in some cases use the top effect condensation enthalpy as part or all of the energy required to heat and melt incoming magnesium, reducing the overall energy required. Third, incoming magnesium scrap can have zinc or other volatile metals with higher vapor pressures than magnesium, which concentrate in the distillate. In addition to the barrier between evaporator and condenser described above, a counter-flow evaporator further segregates zinc into the top condenser; and a vented counter-flow condenser can separate most zinc out of a condenser [13].

Fourth, magnesium alloy evaporation sometimes leads to aerosol creation and carry-over, bringing low-volatility impurities to the condenser. Flow devices which create high velocity and streamline curvature, such as cyclone separators and even angled vapor flow channels, can separate such aerosols and reduce distillate contamination.

Steel Corrosion and Batch Distiller Experiments

Production of magnesium distillate requires the use of nickel-free steels which neither corrode nor contaminate liquid magnesium. For this reason, a 200-hour corrosion test compared five stainless alloys in an inert environment with liquid magnesium: 310, 316L, 347, 410, and 440C. Figure 2 shows the apparatus consisting of a graphite separator in a mild steel container, a welded steel lid and tube for evacuation are not shown. A mechanical vacuum pump reduced air pressure below 10 Pa, then the tube was crimped to seal it. A kiln heated the apparatus to 1100 °C for 200 h. After cooling, cutting open the apparatus revealed the steel coupons embedded in magnesium.

The 3xx series alloys showed considerable corrosion and porosity, with SEM EDS indicating at least 1 wt% Mg in the centers of each of those coupons. Solid Mg discs surrounding the 3xx alloy coupons showed 11–13 wt% Ni. In contrast, 410 and 440C steel samples showed negligible corrosion with no nickel in the magnesium around them. Based on this, the distiller structure should use nickel-free stainless alloys. Strength of these alloys is low, but operating in an inert gas-filled vessel with equal pressure to that of the bottom effect (10–15 kPa) can minimize hoop stress in the hottest part of the distiller, and supports in the lowest effect can be made of ceramic materials.

23 one- and two-effect batch distillation experiments examined behavior of magnesium and a synthetic scrap alloy. A summary of five two-effect experiments follows. Synthetic scrap consisted of AZ31 with addition of ∼4 wt%

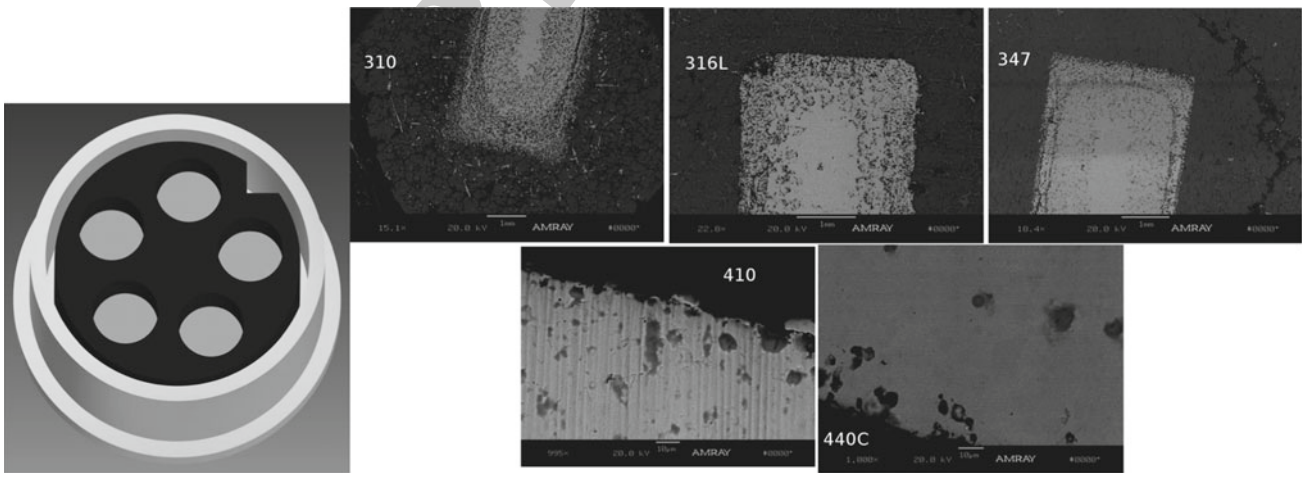


Fig. 2 (Left) mild steel vessel with graphite separator used in corrosion experiments (welded lid not shown); (left) SEM backscatter micrographs of stainless coupon cross section after 200-hr Mg exposure showing alloys 310, 316L, 347, 410 and 440C. Note 1 mm scale bars in 3xx steel samples, and 10 μm scale bars in 4xx samples

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Fe, 1 wt% Cu, and 1 wt% Ni. Side heaters raised the temperature in the whole apparatus to start, these then switched off and only a bottom heater ran during distillation. Most experiments used welded 6-inch schedule 80 pipe and plate with 316 stainless in the bottom evaporator and 1018 steel in the remainder of the apparatus. Figure 3 shows all five setups cut open after the experiments.

Experiment I used pure magnesium in an apparatus made of welded 4-inch schedule 80 stainless pipe cap sections. The top and bottom effects each began with 500 g pure magnesium. The side heater remained on throughout the experiment. All of the magnesium in both effects transferred to the condensers, likely because of the side heater.

Experiment II began with 4.0 kg synthetic scrap in the bottom effect and 2.6 kg in the top. The side heater extended from the bottom of the bottom evaporator to about halfway up the top condenser, leaving the top of the top condenser open. As a result, the top of the top condenser was too cold, so it clogged. Bottom heater power was 2.0 kW, and significant non-volatile impurities entered the bottom condenser.

Experiment III also began with 4.0 and 2.6 kg synthetic scrap in the two effects. The side heater heated the entire apparatus to 650 °C, then shut off. Bottom heater power was only 1.0 kW, and low-volatility impurity concentrations were half or less of those in Experiment II. But in 6.6 h only 2 kg Mg distilled in the bottom effect, and the low resulting heat flux did not distill any of the top effect. Our hypothesis was that at this low power and distillation rate, an Mg-depleted layer formed in the evaporator liquid surface, reducing the evaporation rate.

Experiment IV also began with 4.0 and 2.6 kg of synthetic scrap in the two effects. The side heater heated the apparatus to 650 °C, then shut off, after which the bottom ran at 2.0 kW. Our hypothesis was that as with Experiment II, high heat flux led to bubbling in the bottom evaporator, increasing distillation rate while also creating aerosol carry-over into the condenser. Condensation rate in the bottom effect created enough heat to distill 1.3 kg Mg in the top effect. But with the side heater off during distillation, the tall and narrow apparatus lost a considerable amount of heat to radiation from the sides; this agreed well with reduced-order thermal model predictions described next.

Experiment V began with 3.5 and 2.2 kg of synthetic scrap in the two effects. A piece of angle iron was welded over the bottom effect conduit entrance, requiring vapor to turn 90° twice with just ~1 cm maximum turn radius. The side heater heated the apparatus to 750 °C then shut off, after which the bottom heater ran at 1.5 kW. Distillation was slower than in experiments II and IV, but faster than Experiment III, yet Al, Cu, and Ni in distillate were very low. This supports the aerosol formation hypothesis, with centrifugal separation preventing condenser contamination (Table 1).

Reduced-Order Thermal Model

A reduced-order continuous flow thermal model estimates the temperatures, vapor, and total pressures, heat and mass flow rates, and distillate and concentrate compositions in a multi-effect distiller. It uses the General Solution Model [14] with parameters from Zivkovic for Al–Mg-Zn ternary liquid [15] to estimate vapor pressures. It solves a set of equations simultaneously using a matrix-free Newton–Krylov nonlinear solver.

Fundamentally, distillation rate is limited by heat transfer up through the distiller. This in turn is mostly limited by heat conduction through the steel evaporator support and liquid magnesium in the evaporator. Magnesium vapor flow requires only a very small pressure difference, so within an effect magnesium vapor pressure at the evaporator surface is very close to vapor pressure at the condenser surface—though evaporator temperature is higher due to the lower magnesium activity in liquid solution. This model assumes equal evaporator and condenser vapor pressures.

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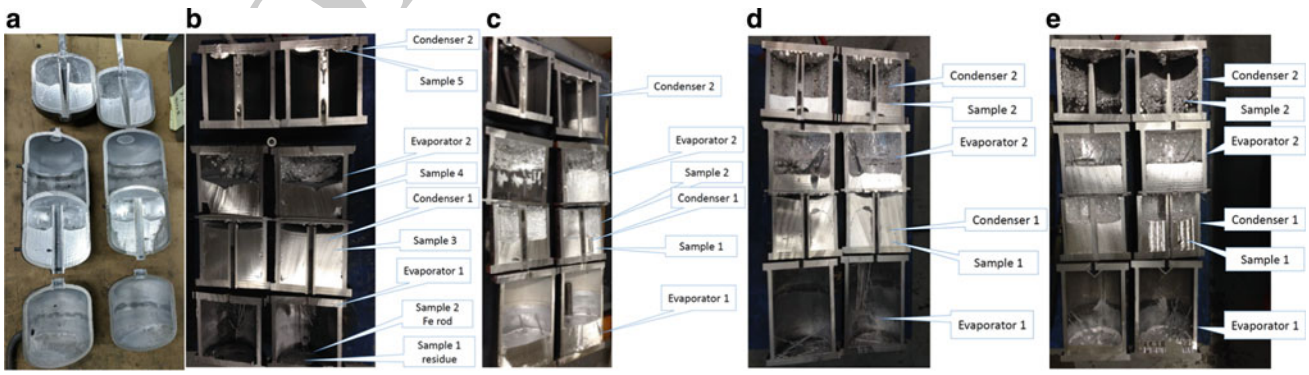


Fig. 3 Two-effect distiller experiment cross sections labeled I, II, III, IV, and V above (left to right), and showing locations of composition analysis samples

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Table 1 Distillation experiment durations, distillation rates, heater power levels, and compositions

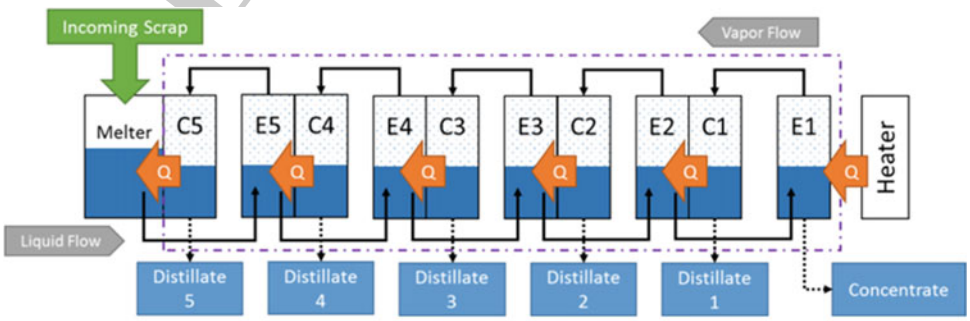
Experiment	I	II*	III	IV	V
Duration at temperature, h	1.5	5.0	6.6	5.0	7.0
Bottom effect rate, kg/h	3	0.8	0.3	0.8	0.5
Top effect rate, kg/h	unk.	0	0	0.25	0
Sample 1 Al wt%		1.3	0.32	0.86	–
Sample 1 Zn wt%		0.8	1.76	1.07	0.012
Sample 1 Cu wt%		0.4	0.13	0.49	–
Sample 1 Ni wt%		0.2	0.12	0.36	–
Sample 2 Al wt%		99	0.01	0.02	
Sample 2 Zn wt%		1	1.33	1.11	
Sample 2 Cu, wt%		–	–	0.05	
Sample 2 Ni, wt%		–	–	0.01	

*Sample 3 and 5 compositions, not 1 and 2. Concentrations below detection limits indicated by "–"

Figure 4 shows a schematic and a result of this thermal model applied to a continuous large laboratory distiller with five effects and total influent flow rate of 2.7 kg/h. Influent composition is Mg-5 wt% Al-1 wt% Zn. Temperatures at the bottom evaporator and top melter are set to 950 °C and 800 °C, respectively. Final concentrate flow rate is set to 10% of influent rate; distillate flow rates in each effect are determined by heat flow balances. Maximum pressure difference between effects is 4 kPa, corresponding to 28 cm of liquid Mg height in a standpipe. Evaporation rate in the higher effects is lower than the lower ones because radiation loss out the sides reduces heat flux up the distiller.

Figure 5 shows results of the same model for 2–7 effect pilot-scale distillers with 1.14 m inner diameter and a target total Mg distillate production rate of 100 kg/hour (500–700 t/a). Bottom evaporator temperature is again 950 °C, and melter temperature is 750 °C. Evaporators are 10 cm high with 2.5 cm liquid Mg depth, and condensers are 17.5 cm high with 0–15 mg liquid Mg depth. Each condenser holds about 230 kg Mg, and must be tapped every 2.5–11 h depending on the number of effects (more effects have lower production rate per effect and need less frequent tapping). When using counter-flow evaporators [13], only the top effect distillate has higher Zn than the influent, production rate of this high-Zn distillate is shown in orange on the stacked graph on the left and the remainder with low Zn is shown in blue. The graph on the right shows that required bottom heater power to achieve these flow rates (shown in

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	Effect 5 C5 E5	Effect 4 C4 E4	Effect 3 C3 E3	Effect 2 C2 E2	Effect 1 C1 E1
Temperature (°C)	810 814	825 831	843 851	864 877	892 950
Pressure (psia)	0.68	0.83	1.00	1.30	1.90

	Distillate 5	Distillate 4	Distillate 3	Distillate 2	Distillate 1	Concentrate
Total (kg/hr)	0.40	0.44	0.49	0.54	0.59	0.27
Magnesium (%)	97.8%	98.3%	98.9%	99.3%	99.7%	50.0%
Zinc (%)	2.2%	1.7%	1.1%	0.7%	0.3%	0.0%
Aluminum (%)	0%	0%	0%	0%	0%	50%

Fig. 4 Five-effect METS schematic, and output of reduced-order thermal model of a 2.4 kg/h large laboratory G-METS magnesium distiller showing distillate flow rates and compositions starting from Mg-5%Al-1%Zn

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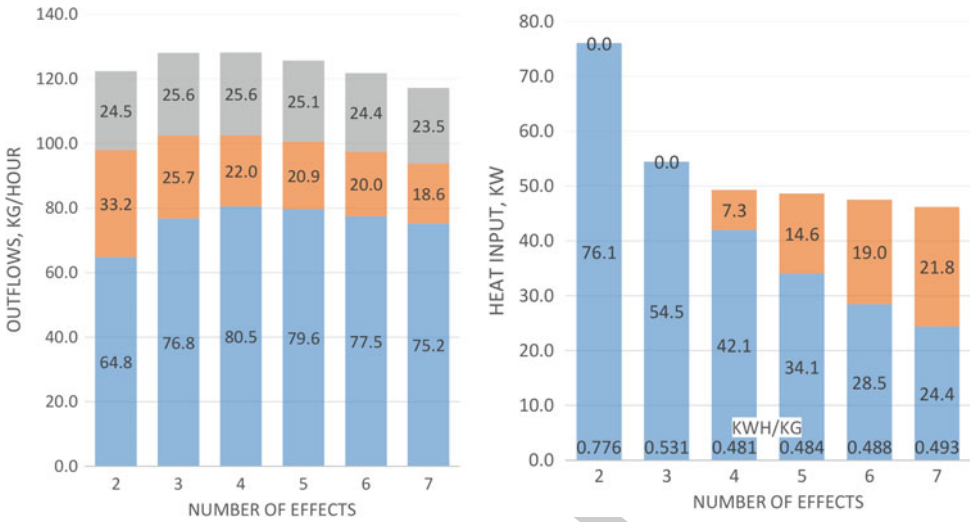


Fig. 5 Reduced-order model estimates of ~100 kg/h pilot distiller outflows (left) consisting of, from the top, concentrate, high-Zn distillate, and <1% Zn distillate; and heat input (right) showing, from the top, melter heater power, bottom heater power, and total specific energy use in kWh per kg of magnesium distillate product

blue) falls with the number of effects, as is expected, but for solid Mg alloy scrap input and more than three effects, heating and melting Mg requires additional heat input to the melter (shown in orange).

Based on the model output, the three-effect distiller was selected for scale-up design and further study, as it is simpler than the larger distillers but with little additional energy input and high-Zn distillate.

First-Cut Industrial Distiller Solid Model and Structural Finite Element Analysis

Figure 6 shows a first-cut scaled G-METS magnesium distiller design with three effects and ~1000 kg/h (5000–7000 t/a) pure Mg production rate. Internal diameter is 3.6 m, and the cross-section area is 10 m^2; this is ten times the area of the 100 kg distiller described above. In the cross section, liquid Mg is shown in dark gray, with a thin 2.5 cm layer in each evaporator and up to 15 cm of distillate storage (~2300 kg) in each condenser. There is a vessel at the bottom where concentrate accumulates. Tapping can be done by melting frozen plugs in the drain pipes.

The outer shell maintains inert gas pressure roughly equal to Mg vapor pressure in the lowest evaporator in order to minimize hoop stresses in that hottest part of the distiller. Both 410 and 440 nickel-free stainless alloys were found to exhibit negligible corrosion in experiments which exposed them to Mg liquid and vapor for 500 h, but both are relatively weak at high temperature. The structural Finite Element Analysis shown in Fig. 7 with gravitational and pressure forcing shows stress in the bottom effect supports around 5–8 MPa, with maximum stress in a few spots up over 27 MPa. It may be necessary to make some of these out of a non-reactive ceramic material with high compressive strength at 950 °C such as AlN, MgO, or CaO. Higher effects will have additional inward hoop stresses due to lower internal pressure, but those regions will be at a lower temperature such that steel will be stronger.

Flow and Heat Transfer Finite Element Analysis

In this study of a G-METS magnesium distiller, FEA of flow and heat transfer has two major goals: model heat conduction in steel and magnesium which limits the performance of the distiller, and model coupled evaporation and flow in gas phases for later use to design counter-flow evaporator and condensers.

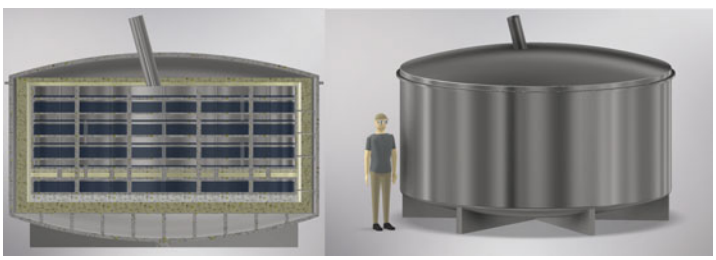


Fig. 6 1000 kg/hour G-METS magnesium distiller design cross section (left) and approximate scale (right)

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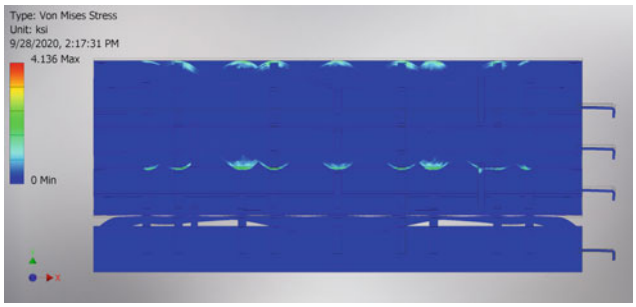


Fig. 7 Structure FEA results showing stresses ~2 ksi = 13.8 MPa in lower supports, and 28 MPa max stress

This model uses Elmer Version 8.4 with the Heat Model and Navier-Stokes Model for gas phase fluid flow. The geometry is based on the 1.14 m inner diameter ~100 kg/hour 3-effect distiller and meshes were made with Salome Program version 7.7.1 and the NETGEN 1D-2D-3D algorithm. The mesh has 92.480 nodes with element size from 0.001 to 0.05 m. In this geometry, each effect includes steel as a body of the bottom effect, liquid magnesium in the evaporator, steel as body between evaporator and condenser, and a condenser boundary condition on top of the top of effect. Vapor pressure is calculated from temperature using the Clausius–Clapeyron equation with liquid Mg constants [16] and the heat flux boundary condition at the liquid-gas interface of liquid and gas magnesium includes evaporation enthalpy. The model uses temperature-dependent thermal conductivities [17], viscosities [18, 19] and densities of magnesium in the liquid and vapor phases.

Figure 8 shows the distiller model geometry, velocity magnitude throughout the distiller, and velocity and relative pressure along the vertical axis of the vapor conduits. Maximum velocity is about 8 m/s, leading to a Reynolds number in the conduit around 250, so these simulations ran without a turbulence model.

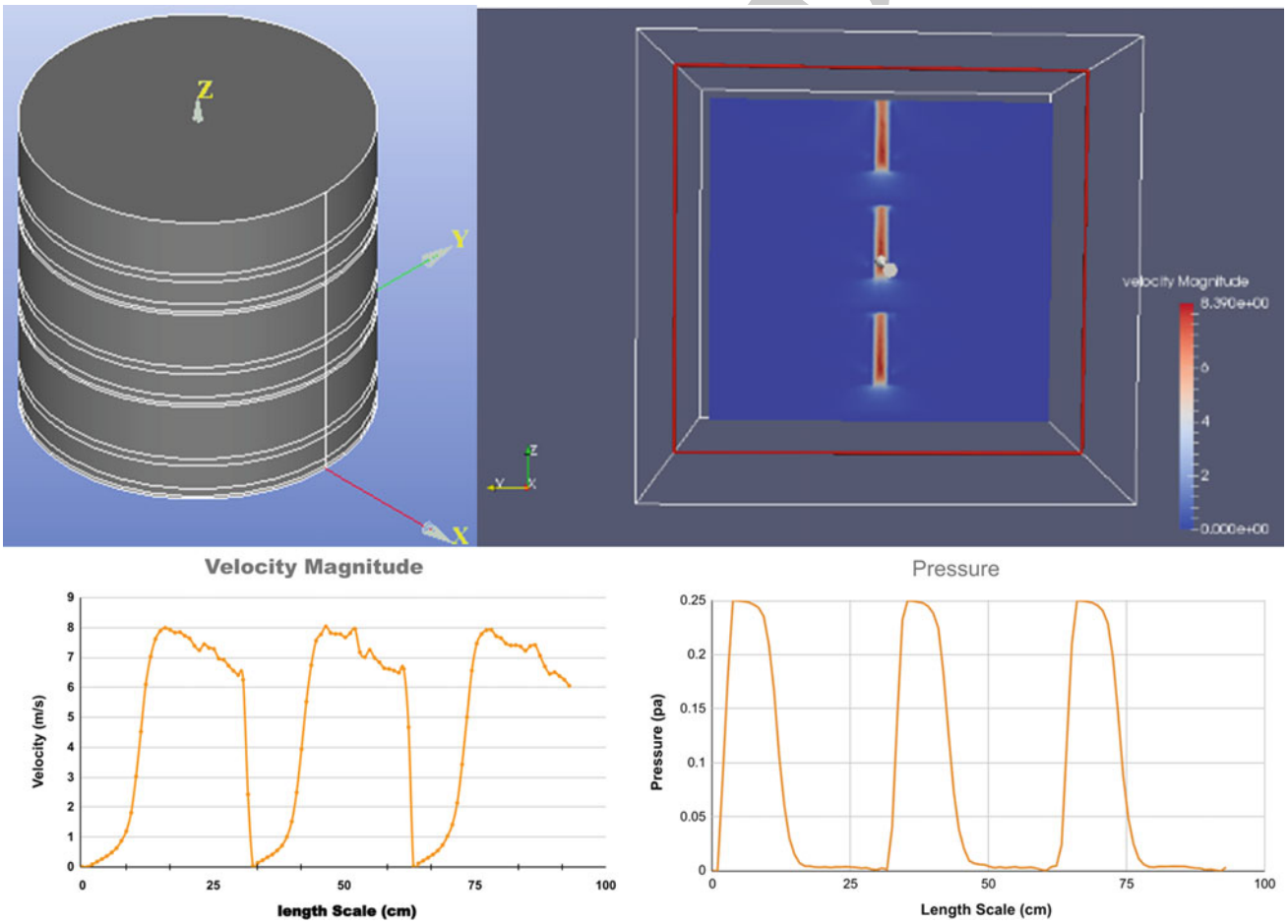


Fig. 8 FEA model geometry, cross-section velocity magnitude, and center axis velocity and pressure profiles

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Discussion

The biggest potential issue with the G-METS distiller is the kinetics of alloy evaporation. Experimental work showed that although pure Mg distillation rate very closely followed the reduced-order thermal model predictions, when an alloy was used, the rate was either much slower than the pure Mg model predicted with low heat flux, or with sufficient heat flux, matched the pure Mg distiller—but with product contamination by less volatile impurities such as Al. The current hypothesis for this behavior is formation of a surface layer depleted of magnesium which impedes evaporation at low heat flux, and disruption of that layer by bubbling with aerosol creation at higher heat flux. Though early experiments have supported this hypothesis as described above, characterization of this phenomenon requires further study, as does its remedy by cyclone or other flow curvature separation.

Conclusions

This work presented: results from batch magnesium distillation experiments, a validated reduced-order thermal model applied to a pilot-scale 100 kg/h distiller, a 1000 kg/h distiller design study with an estimation of stress due to weight of the distiller and pressure difference, a thermal-fluid FEA model of vapor flow, and techno-economic analysis. These show that a G-METS magnesium distiller can potentially reduce the energy use from 5 to 7 kWh/kg today to as little as 0.5–1 kWh/kg, a nearly 90% reduction. The distiller is continuous and compact, and can potentially enable new recycling processes for low-grade post-consumer magnesium alloy scrap. Further study is needed to clarify the kinetics of alloy evaporation, to validate the reduced-order and FEA thermal-fluid models, to identify potential limiting mechanisms in scale-up, and to engineer key subsystems in the distiller.

If it succeeds, this distiller could enable electrolytic production of magnesium using a Hall–Héroult type cell with reactive dense liquid metal cathode e.g., tin, as first envisioned in the 1940s. The lower valence of Mg than Al leads to higher productivity in a plant with the same potline current, potentially producing Mg at or below the cost of Al.

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